

Ian Burke

Software Engineer · AI Integration & Automation

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Build and maintain AI products, data pipelines, and tools for working with coding agents. Work across user interfaces, backend services, provider integrations, and production troubleshooting. Background in commercial video production.

EXPERIENCE

Davenport University

Current

Automation & AI Integration Specialist · Web Development and Custom Applications

Design and ship AI-backed tools for university staff, from a Google Chat assistant to automated donor research briefings. Own the full path from prototype to production on university infrastructure: deployments, CI/CD on hosts with no inbound access, per-app model keys with spend budgets, and deploy gates that verify each shipped capability actually works.

- Built "Pounce," a Google Chat AI assistant for university staff (Next.js, SQLite, Gemini and on-prem Gemma/Llama via a LiteLLM gateway), with slash commands for status reports, donor briefings, and bug/idea intake.
- Designed the CI/CD pattern for the university's AI host: self-hosted pull-based deploys with systemd and Docker Compose, gated by a capability-level post-deploy verifier.
- Automated donor research briefings (Blackbaud NXT plus grounded web research) delivered as Google Docs from Chat, replacing manual prep.
- Stood up a shared LiteLLM gateway with per-app virtual keys, budgets, and a five-node on-prem Gemma pool.

Narrative Ads

2022 - Present

Freelance Software Engineer

- Build and maintain NarratorAI, an internal platform for AI media production and advertising analytics, using TypeScript, Next.js, and Firebase.
- Develop image, video, and voiceover workflows with Gemini, fal.ai, and ElevenLabs, alongside advertising-data integrations.
- Investigate generation failures, deploy backend capacity changes, and reconcile provider billing into monthly infrastructure reports.

Narrative Ads

2018 - 2022

Video Producer & Editor

Directed, produced, and edited paid social ads for brands including DoorDash, Harry's, and Native. Managed creative direction, revisions, and production handoffs.

SELECTED PROJECTS

Kill Tony Archive · AI data pipeline and web application

Built separate transcription and structured-extraction stages for long recordings, with transcript deduplication and speaker normalization. The September 2026 local dataset contains 1,422 set records across 110 episodes. Python, Gemini, SQLite, FastAPI, and SvelteKit.

PLEXI · Desktop workspace for coding agents

Built a Rust application combining terminal panes, project contexts, notes, and apps. Includes an agent-accessible CLI and a WASM app runtime. Used as my daily working environment; macOS pre-release.

TECHNICAL SKILLS

Python, TypeScript, JavaScript, Rust · React / Next.js, SvelteKit, FastAPI, Node.js · Firebase, Google Cloud, SQLite, GitHub Actions · AI provider integration, structured extraction, agent workflows, incident investigation